

Floating offshore wind farm fault detection and isolation benchmark

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2. Description

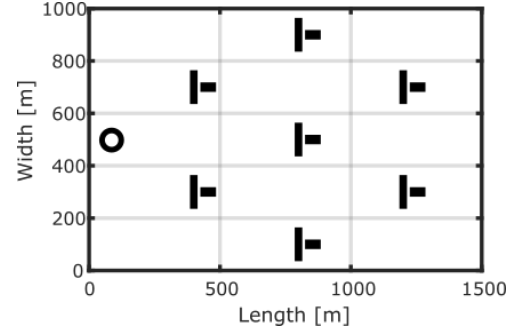
Detecting faults and ensuring reliable control of wind turbines, even under faulty conditions, are essential for reducing the cost of wind energy and increasing a robust integration of the generated power into electrical grids. Improvements in fault detection and isolation (FDI) and fault-tolerant control (FTC) can significantly enhance the reliability and operational efficiency of wind turbines, particularly as they are deployed in remote and less accessible offshore locations. In recent years, research on wind turbine FDI and FTC has surged, with studies exploring various techniques to detect and mitigate faults in critical components such as blades, pitch actuators, generators, and power converters. As interest in this field has grown, the need for standardized benchmark datasets has become evident [1-3], facilitating research and enabling fair comparisons among different approaches. In response to this need, several benchmarks have been proposed, mostly targeting onshore wind turbines.

However, as the industry moves toward larger turbines located farther offshore—requiring floating platforms—existing benchmarks no longer fully capture the evolving challenges. This motivates the need for a new benchmark that better reflects the current state of offshore wind technology. Thus, a new benchmark is proposed here that differs from previous ones in several key aspects:

- **Environment:** Developed using the FOWLTy simulator [4], this benchmark provides an easy-to-understand, Simulink-based coding environment, making it accessible to researchers without a strong background in wind energy.
- **Floating offshore wind farm:** The test case consists of a wind farm with seven floating offshore wind turbines (NREL's 5MW turbine [5] on a DeepCWind platform [6]).
- **Diverse wind conditions:** Instead of relying on a single wind file (with a fixed or variable mean speed), this benchmark includes ten different simulations with varying wind speeds, covering different operational ranges.
- **Fault severity variability:** By leveraging multiple simulations, the severity of faults is varied across wind scenarios, enabling a more comprehensive evaluation of FDI strategies and their sensitivity to different fault magnitudes.
- **Customizable training data:** Since the benchmark is based on a simulator, researchers can generate additional wind profiles and operational data to train and validate data-driven models for FDI and FTC.

3. Data

The benchmark considers a wind farm composed of seven floating wind turbines and a measuring mast, arranged as shown in Figure 1. To facilitate FDI strategies development, the benchmark provides access to a set of sensor measurements and internal reference values for the actuators. To enhance realism, artificial measurement noise has been added to all sensor-provided variables, with the corresponding noise power values [3] indicated in parentheses. The available variables are:



Sensor measurements:

- Rotor rotational speed (10^{-4} rad/s)
- Generator rotational speed ($2 \cdot 10^{-4}$ rad/s)
- Generator torque (0.9 Nm)
- Generator power (10 W)
- Wind speed at the nacelles and measuring masts (0.0071 m/s)
- Blade pitch angle ($1.5 \cdot 10^{-3}$ deg)
- Nacelle acceleration ($5 \cdot 10^{-4}$ m/s²)

Reference signals (internal actuator commands):

- Blade pitch angle reference
- Generator torque reference
- Farm-level demanded power

4. Fault scenarios

This benchmark provides ten distinct datasets, each representing a different operational condition of the wind farm. These conditions are characterized by varying mean wind speeds (from 5 m/s to 23 m/s), with each dataset containing a unique set of faults. Faults occur one at a time and affect only a single wind turbine at a time, with durations ranging from ten to fifty seconds, depending on the specific case. Additionally, both the severity and timing of faults vary across datasets, ensuring diverse fault scenarios for evaluation. The faults considered in this benchmark (and their maximum severity values) have been selected following common faults in the literature [1,2,7,8] and are categorized into **sensor faults** and **actuator faults**, as detailed below:

Sensor Faults

- **Pitch Sensor Faults:** Pitch sensors provide critical feedback for controlling blade angles. The considered faults include (i) a **stuck sensor**, where the measurement remains frozen due to electronic failure or mechanical blockage; (ii) a **scaling fault**, where the sensor output is scaled incorrectly due to calibration errors or component degradation (20% max); and (iii) a **drifting sensor**, where the measurement gradually shifts over time due to thermal effects or sensor wear (1 deg/s max).
- **Generator Sensors Faults:** Generator sensors monitor power production and rotational speed. The faults include (i) **power scaling**, where incorrect sensor calibration or interference causes the power measurement to be scaled incorrectly (10% max), and (ii) **speed scaling**, where signal processing errors or encoder degradation result in an inaccurate speed reading (5% max).

Actuator Faults

- **Pitch Actuator Faults:** Pitch actuators adjust blade angles to optimize power output. The considered faults include (i) a **stuck actuator**, where mechanical failure or hydraulic leaks

lock the actuator in place; (ii) an **offset fault**, where calibration errors or incorrect initialization cause a persistent misalignment (3 deg max); and (iii) an actuator gain fault introducing a multiplicative scaling error to the commanded pitch angle (10% max), representing internal position feedback miscalibration or proportional pressure anomalies in the hydraulic system.

- **Generator Actuator Faults:** Generator actuators regulate the torque applied to the generator. The main fault considered is a **torque offset**, where incorrect sensor calibration or power electronics aging leads to a persistent deviation in commanded generator torque (1 kNm max).

5. Evaluation criteria

To ensure consistency and facilitate comparison across different FDI strategies, a set of evaluation criteria is defined for researchers to follow when assessing their methods. The proposed criteria consider key aspects of fault detection, isolation, and computational efficiency.

Fault Detection Performance

- **False Alarm Rate (FAR):** The proportion of times a fault is falsely detected when no fault is present. A lower FAR indicates a more reliable detection system.
- **Missed Detection Rate (MDR):** The proportion of actual faults that go undetected. A lower MDR reflects higher sensitivity to faults.
- **Detection Time:** The time elapsed between the fault occurrence and its detection. While faster detection is preferable, it must be balanced with reliability to avoid false alarms.

Fault Isolation Accuracy

- **Correct Isolation Rate:** The percentage of detected faults that are correctly identified, ensuring that the diagnosed fault type matches the actual fault.
- **Ranking of True Fault:** If the method provides a ranked list of possible faults, the position of the correct fault in this ranking serves as an evaluation metric.

Computational Efficiency

- **Computation Time:** The time required for the detection and isolation algorithm to process the data. To account for differences in computational power, this value should also be reported **normalized** against the time required to run one of the datasets (with 11 m/s wind speed) using the benchmark model without an FDI strategy.

By following these evaluation criteria, results from different studies can be made comparable, ensuring fair benchmarking of FDI methodologies.

6. Resources

The benchmark provides two types of resources, both available in the downloaded “**FDI Benchmark**” folder (which can be obtained [here](#)):

- **Simulink model** and supporting files for reproducing and extending the simulations.
- **Precomputed datasets** of simulated variables for offline FDI/FTC development directly in MATLAB.

The files included in both types of resources are explained below.

Simulink model

To replicate the benchmark conditions and generate new simulations, users must install the **FOWLTY toolbox**, which is available [here](#). Once FOWLTY is installed, the benchmark data and necessary files can be found in the “FDI Benchmark” folder within the downloaded files. The provided resources include:

- **Simulink Model:** A file containing the offshore floating wind farm model used for the simulations.
- **MATLAB Initialization File:** This file allows users to select from the predefined wind datasets, each corresponding to a specific operational condition and fault scenario.
- **Wind files:** The wind field files defining the wind speeds for all the considered data sets. They can be found inside the “data” folder.
- **Wind Dataset Generation File:** An additional MATLAB script enabling the generation of new wind conditions, allowing users to expand the dataset for training and testing purposes.

These resources ensure that researchers can replicate the benchmark conditions while also generating new scenarios to test the robustness of their FDI strategies. For a detailed description of the **FOWLTY simulator**, interested readers are referred to [4].

Precomputed datasets

The folder also includes **precomputed datasets of the simulated variables**. These contain the sensor measurements, actuator references, and fault scenarios described earlier, stored in MATLAB-readable format.

This option allows researchers to perform **offline FDI and FTC development directly in MATLAB**, without running Simulink. It is particularly useful for rapid prototyping, data-driven approaches, and for those without access to Simulink licenses.

7. References

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